

Small Intestine

The small intestine or small bowel is the part of the gastrointestinal tract between the stomach and the large intestine, and is where most of the end absorption of food takes place. The small intestine has three distinct regions – the duodenum, jejunum, and ileum. The duodenum is the shortest part of the small intestine and is where preparation for absorption begins. It also receives bile and pancreatic juice through the pancreatic duct. The primary function of the small intestine is the absorption of nutrients and minerals from food, using small finger-like protrusions called villi. Of note: Leonardo Da Vinci mistakenly believed the small intestines help with breathing by storing air and pushing the diaphragm upward!

The Small Intestine: A Health Context

Gastroenteritis, also known as infectious diarrhea, is inflammation of the gastrointestinal tract—the stomach and small intestine. Symptoms may include diarrhea, vomiting, and abdominal pain. It is not related to influenza though it has been called the "stomach flu". It is also known as "food poisoning" when caused by consuming contaminated food or drink.



If you have experienced gastroenteritis, you might recall being advised to avoid milk products during your recovery. This is because gastroenteritis can cause temporary lactose intolerance (lactose is a sugar found in milk). Gastroenteritis damages the simple columnar epithelium in the small intestine and prevents the brush border from producing lactase, the enzyme that digests lactose. Some studies have shown lactase is most strongly produced in the mid-regions of the small intestine and less so around the connections to the stomach and large intestine. It is those regions that no longer allow the proper processing of lactose after gastroenteritis.

Liver

The liver is a reddish-brown, wedge-shaped organ with four lobes of unequal size and shape. A human liver normally weighs approximately 1.5 kg (3.3 lb), and has a width of about 15 cm (6 in). It is both the heaviest internal organ and the largest gland in the human body. Located in the right upper quadrant of the abdominal cavity, it rests just below the diaphragm, to the right of the stomach and overlies the gallbladder. The falciform ligament attaches the liver to the anterior (ventral) body wall, and separates the liver into the right lobe and left lobe. The ligament stretches from the underside of the diaphragm to the posterior surface of the sheath of the right rectus abdominis muscle. The liver is connected to two large blood vessels: the hepatic artery and the portal vein and common hepatic duct. The hepatic artery carries oxygen-rich blood from the aorta via the celiac plexus, whereas the portal vein carries blood rich in digested nutrients from the entire gastrointestinal tract and also from the spleen and pancreas. These blood vessels subdivide into small capillaries known as liver sinusoids, which then lead to lobules. Lobules are the functional units of the liver. Each lobule is made up of millions of hepatic cells (hepatocytes).

The gallbladder is a small hollow organ where bile is stored and concentrated before it is released into the small intestine. The pear-shaped gallbladder lies beneath the liver. It receives and stores bile, produced by the liver, via the common hepatic duct and releases it via the common bile duct into the duodenum, where the bile helps in the digestion of fats.

The Liver and Gall Bladder: Two Health Contexts



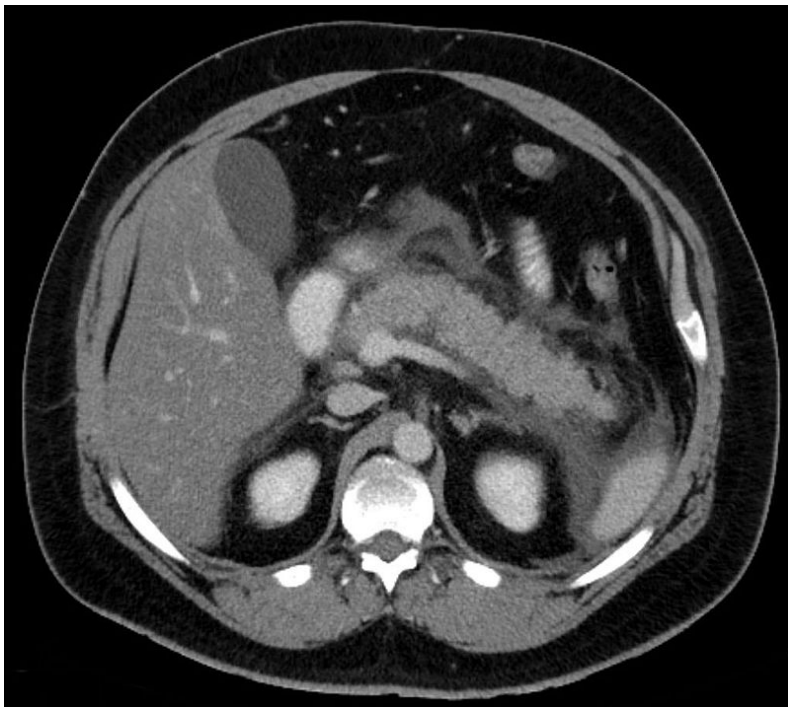
Jaundice is a condition where bilirubin (a waste product from the breakdown of hemoglobin) builds up in the blood. This causes the skin to look yellow and can lead to serious complications, or even death. Normally, bilirubin exits the body in bile. Hepatocytes use bilirubin to make bile, then excrete the bile via the bile duct. But occasionally gallstones block the bile duct, causing a backup of bile into the liver (shown on CAT scan). Bile then enters the blood, leading to jaundice.



Codeine is an opioid pain medication, found in some prescription cough syrups. Codeine itself is not active as a pain reliever, until it is converted to morphine by hepatocytes. Codeine is easily absorbed in the intestines, where it then goes to the liver and is converted to morphine before circulating through the rest of the body.

Pancreas

The pancreas is an organ of the digestive system and endocrine system of vertebrates. In humans, it is located in the abdominal cavity behind the stomach. The pancreas is a mixed gland, having both an endocrine and an exocrine function. As an endocrine gland, it secretes into the blood several important hormones, including insulin, glucagon, somatostatin, and pancreatic polypeptide. As an exocrine gland, it secretes pancreatic juice into the duodenum through the pancreatic duct. This juice contains bicarbonate, which neutralizes acid entering the duodenum from the stomach; and digestive enzymes, which break down carbohydrates, proteins, and lipids in ingested food entering the duodenum from the stomach. Of note: About 20,000 protein coding genes are expressed in human cells and about 50% of these genes are expressed in the normal pancreas. Also of note: Early medical experts found it quite difficult to figure out the function of the pancreas. For a long time, because of its rubbery texture, anatomists believed the pancreas was nothing more than a shock absorber to cushion the stomach against blows.



The Pancreas: A Health Context

Inflammation of the pancreas (shown in CT scan to left) is known as pancreatitis. Pancreatitis is most often associated with recurrent gallstones or chronic alcohol use, although a variety of other causes, including measles, mumps, some medications, and even some scorpion stings, may cause pancreatitis. Pancreatitis is likely to cause intense pain in the central abdomen, that often radiates to the back, and may be associated with jaundice. One cause of jaundice is the inability of bile to escape the gall bladder, causing bile to back up and enter the blood (more on that in Station 8).

